Monitoring and improving the FAIRness of EarthPortal semantic artefacts

First conclusions: five notebooks, one audit per ontology, and what we learned about O'FAIRe

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CENTRE NATIONAL DE LA RECHERCHE SCIENTIFIQUE

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Today in 30 minutes

1. **Where I am** in the internship
2. **O'FAIRe in five minutes**: what it measures, and how we read it
3. **The pipeline**: five notebooks, from the API to one audit per ontology
4. **What we found**: in the catalogue, and about O'FAIRe, with evidence
5. **Environment and next steps**

The one message of this talk

We use O'FAIRe as a **measuring instrument** to monitor and improve the FAIRness of the artefacts hosted on EarthPortal.

We do **not** change its implementation: improving O'FAIRe is another project, for someone else. But to read its scores correctly, we had to understand exactly how it computes them.

Where I am in the internship

Task	What exists today	Status
1. Metadata guideline	Quick reference table (cheatsheet) of the 99 attributes, typos corrected (Report 1); a **machine-readable** version generated from it	first version
2. Audit of all artefacts	92 audits, one per catalogue entry, with ranked actions	first run
3. Towards automation	5 notebooks, reproducible on any dated snapshot of the catalogue	manual run
4. FAIR metrics	O'FAIRe questions, properties and Java implementation documented (Reports 2 to 4); FOOPS! not studied yet	in progress

All code, tables, audits and reports: github.com/EarthPortal/SA_Management (folders `docs/` and `earthportal_sa_audit/`).

EarthPortal on 8 October 2026, in numbers

92
public catalogue entries

- **68** artefacts
- **19** views
- **5** metadata-only records

130
metadata fields stored by the portal

- 110 on each submission
- 20 on the ontology record

99
attributes in the Metadata guideline

- 20 Mandatory
- 58 Recommended
- 21 Optional

The catalogue moves: why every result carries a date

Between 6 and 8 October, `OSO-DCAT` and `OSO-VOID` became **views of OSO**, `OSO-SHACL` appeared as a new view, and `SCHEMA-ORG` left the public catalogue (71 / 16 / 5 became 68 / 19 / 5). Our pipeline works on **dated snapshots**, so each run can be reproduced and compared.

O’FAIRe in five minutes

What is O'FAIRe?

O'FAIRe (*Ontology FAIRness Evaluator*) scores how FAIR a semantic artefact is, from **its metadata record** on an OntoPortal instance (AgroPortal, EarthPortal, ...).

- 4 principles → 15 sub-principles → **61 questions**
- each question pays credits on a ladder, from 0 to its maximum m_q
- fully automatic: the metadata held by the portal, plus a few live checks (do the URLs resolve?)

	F	A	I	R	**Total**
Questions	13	13	15	20	**61**
Maximum	113	113	109	143	**478**

The score of an artefact o

$$S(o) = \sum_{q=1}^{61} s_q(o), \qquad 0 \le s_q(o) \le m_q$$

$$N(o) = \left\lfloor 100 \cdot \frac{S(o)}{478} \right\rfloor$$

478: the sum of the maxima of the 15 sub-principles.

Example, OSO: $S = 364.36$, so $N = \lfloor 76.2 \rfloor = 76$.

Amdouni, Bouazzouni & Jonquet (2022), *O'FAIRe makes you an offer: metadata-based automatic FAIRness assessment for ontologies and semantic resources*, IJMSO 16(1).

How we use it: an instrument, not an object of study

What we do

- take O'FAIRe's answers **as they are**, for every entry of the catalogue
- link each question to the guideline attributes it reads
- turn the lost points into **curation actions** for authors, curators and the portal
- follow the scores over time, snapshot after snapshot

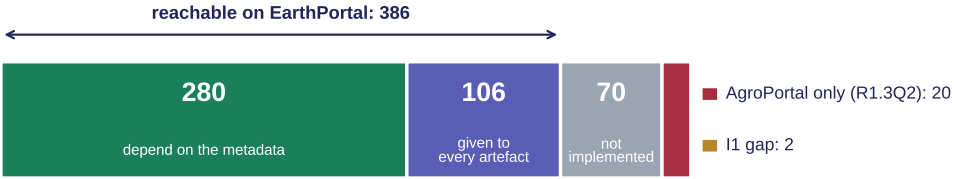
What we do not do

Modify, fork or re-implement O'FAIRe. Its improvement is a separate project.

But we had to read the code

To tell a real 0 from a question that measures nothing, we read the Java source (`agroportal/fairness`, commit `24def6b`, 4 June 2026) and checked every claim on the EarthPortal data.

What the 478 points measure on EarthPortal



Only **386** points can be earned by an EarthPortal artefact. Beside O'FAIRe's own score, we report a **reachable score**:

$$R(o) = 100 \cdot \frac{S(o)}{386}$$

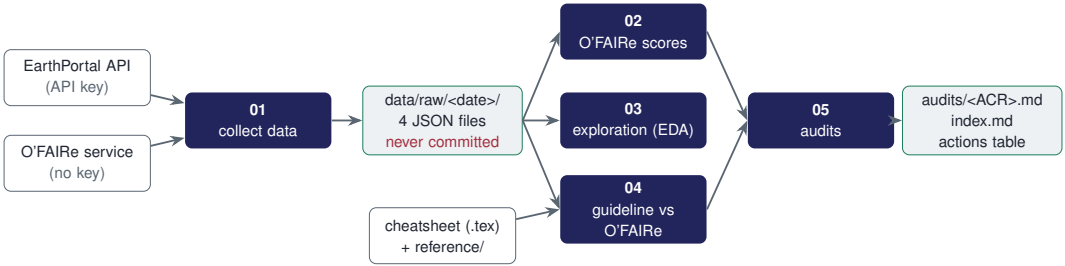
It is fair: every removed question gives 0 to everyone, so $S(o)$ only holds reachable points.

OSO	O'FAIRe N	Reachable R
	76 / 100	**94.4 %**

Split read from the Java code: 12 questions are always given in full (106 points), 10 are not implemented (70).

The pipeline: five notebooks

Overview: one door to the outside world, then pure computation



- **Only notebook 01 talks to the network.** Notebooks 02 to 05 read a frozen snapshot: the same date always gives the same tables, even if EarthPortal changes or is offline.
- One `config.py` (paths, snapshot date) and one `.env` (API key) for all notebooks.

Notebook 01: collecting the data

Part	API call	Saved as	8 Oct
I. Catalogue	`/ontologies` (all fields, views included)	`catalogue.json`	92 entries
II. Portal schema	`/submission_metadata`, `/ontology_metadata`	`metadata_schema.json`	110 + 20 fields
III. Metadata	`/ontologies/<ACR>/latest_submission`	`submissions.json`	92
IV. O'FAIRe	`ofaire/?portal=earthportal`, 10 at a time	`ofaire.json`	92 results

Robust by design

The key travels in a header, never in a URL; timeouts and three tries before failing; answers are saved **untouched**, in a folder named after the snapshot date.

Why data/raw/ is never committed

The raw answers contain contact e-mail addresses, and some pull URLs embed API keys of *other* OntoPortal instances.

Notebook 02: from a nested tree to one table

Each O'FAIRe result is a tree: *ontology* $\rightarrow$ *principle* $\rightarrow$ *sub-principle* $\rightarrow$ *question*. We flatten it into **one row per ontology** (92 $\times$ 68 columns), and we keep **a missing answer apart from a real 0**:

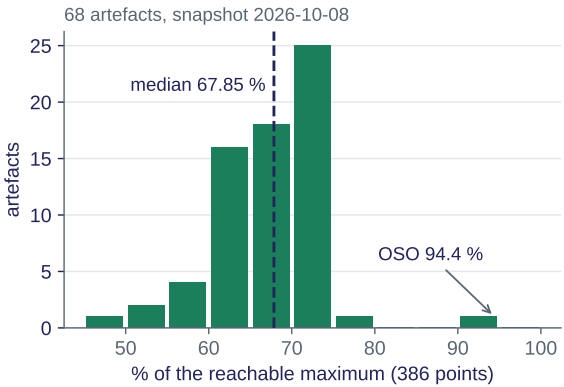
Questions	#	Stored as
Normal	49	O'FAIRe's score
Not implemented	10	empty (`NaN`)
F3Q1, always 0	1	empty (`NaN`)
R1.3Q2, AgroPortal only	1	its real 0, out of the 386

Three checks against O'FAIRe itself

1. the 61 question scores add up to O'FAIRe's total, for **92** / **92**
2. $N = \lfloor 100\, S/478 \rfloor$ holds for **92** / **92**
3. $386 = 478 - 90 - 2$ agrees with the definition

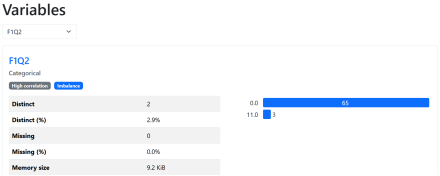
The rules for F3Q1 and R1.3Q2 live in `reference/ofaire_known_issues.csv`, each with its evidence (Java file, line, commit), not hidden in the code.

Notebook 03: how FAIR is the catalogue today?

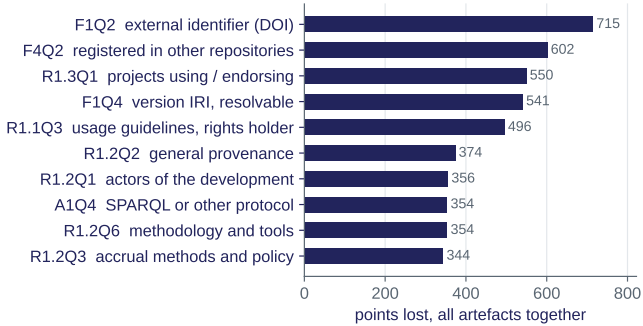


68 artefacts, 49 questions they can score on:

- reachable score: median **67.85 %**, from 49.5 % (STY) to 94.4 % (OSO)
- O'FAIRe's own score: median **54** / **100**
- one clear leader; most artefacts between 60 and 75 %



Where the catalogue loses most points



Points lost $= \sum_o (m_q - s_q(o))$ over the 68 artefacts.

- **F1Q2**: 65 artefacts have no external identifier; only a valid **DOI** gives the full 11 points
- **F1Q4**: 59 have no version IRI
- **R1.1Q3**: nobody is at 0, but almost everybody loses part of the 15 points

These first rows are where curation pays most.

Notebook 04, step 1: the guideline becomes machine-readable

(a) Read the cheatsheet (Report 1, LaTeX) with a regular expression: one table row gives one attribute.

```
1.7 & Status & M & R & A2 & \propb{omv:status} \\
```

→ id 1.7 | Status | Mandatory | FAIR A2 | `omv:status`

(b) Find where EarthPortal really stores it: the first rule that works.

Rule	How	Attributes
same	the guideline property is the portal's own	72
mapping	in the portal's own `metadataMappings`	21
manual	`reference/manual_links.csv`, with reasons	6
	linked	**99** / **99**

26 attributes are stored under another property, e.g.

`mod:acronym` → `omv:acronym`
`dcat:keyword` → `omv:keywords`
`omv:versionIRI` → `owl:versionIRI`

(c) Add what a program needs: the kind of value the portal expects, its allowed values, and **who can fill it in** (author 86, curator 8, portal 5: a proposal to validate together).

⇒ `mapping/guideline_profile.csv` and `.json`: **the machine-readable guideline**.

Notebook 04, step 2: which guideline attributes does each question read?



A trap we found

The `properties` listed in O'FAIRe's configuration are **display only**; the Java code decides what is read. Where they differ, we follow the code (`reference/ofaire_code_vs_config.csv`, 4 rows):

F2Q1 `+schema:translator`; I3Q3 `+dct:language`; I3Q3 `−dct:subject`; R1.3Q2 `+omv:acronym`

Result

(`mapping/ofaire_vs_guideline.csv`):

- 61 questions, **158** properties read
- 33 have no cheatsheet entry: portal metrics and links, object description properties, version information, `owl:deprecated`

Notebook 05: how one file per ontology is written

For each of the 92 entries:

1. **lost points** on each reachable question: $L_q = m_q - s_q(o)$
2. **empty attributes** of the guideline: null, `""`, `[]`, `{}`, or an agent with nothing but its identifier
3. each empty attribute a gets a **share** of the lost points:

$$\text{share}(a) = \sum_{q \text{ reads } a} \frac{L_q}{n_q}$$

   n_q: number of empty attributes that q reads. A ranking aid, not a promise of points.

4. a **priority**, then one Markdown file: `audits/<ACRONYM>.md`

Priorities	
P1	Mandatory
P2	Recommended, costs points
P3	Recommended only
P4	Optional, costs points

Common sense rules: *view of* is asked only of views; *was invalidated by* only of retired artefacts.

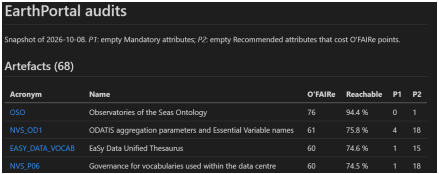
Also written: `audits/index.md` (all entries, best first) and `data/actions_<date>.csv` (5 552 actions).

Example: the audit of OSO, the best artefact

OSO: Observatories of the Seas Ontology
76 / 100 | 364.36 / 478 | **94.4 % of 386** | rank 1 of 68

Question	Points	Lost	O'FAIRe says
F4Q2	2 / 10	8	registered in 1 ontology repository
A1Q3	18 / 24	6	6 negotiation formats supported
I1Q5	0 / 4	4	no other formats declared
R1.2Q2	4 / 6	2	2 provenance properties
I2Q7	3.36 / 5	1.64	metadata vocabularies reused

Suggested action, by the author:
[P2] Fill **Is format of** (cheatsheet 6.2, `dct:isFormatOf`); share of the points lost: 4 (I1Q5)


EarthPortal audits

Snapshot of 2026-10-08. *P1*: empty Mandatory attributes; *P2*: empty Recommended attributes that cost O'FAIRe points.

Artefacts (68)

Acronym	Name	O'FAIRe	Reachable	P1	P2
OSO	Observatories of the Seas Ontology	76	94.4 %	0	1
NVS_OD1	ODATIS aggregation parameters and Essential Variable names	61	75.8 %	4	18
EASY_DATA_VOCAB	EaSy Data Unified Thesaurus	60	74.6 %	1	15
NVS_P06	Governance for vocabularies used within the data centre	60	74.5 %	1	18

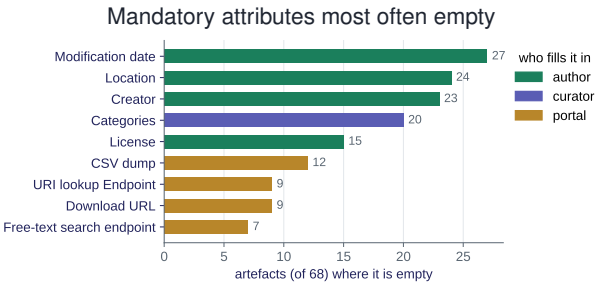
The actions, and who can take them

Artefacts only, priorities 1 and 2:

Who	P1	P2	All
author	89	1 819	1 908
curator	20	61	81
portal	37	0	37
All	**146**	**1 880**	**2 026**

Location (pull URL) is empty *by design* when a file is uploaded: should it stay Mandatory?

Portal rows (CSV dump, download URL, endpoints) point to processing that did not produce a value.



What we found

Reading O'FAIRe scores correctly: six facts, with evidence

Observation	Effect on EarthPortal (8 Oct)	Evidence
10 questions are not implemented	0 for 92 / 92 entries, 70 points; always "cannot be evaluated"	setNotResolvable; our check on 5 612 answers
F3Q1 is a stub; F3Q2 and F3Q3 are always given	F3 reads 21 / 21 for everyone: it carries no information	F3.java, line 25
R1.3Q2 rewards hard-coded group names (OBO-FOUNDRY, WHEAT, CROP, INRAE)	0 / 20 for 92 / 92; none of our 15 groups matches	R13.java, lines 50 to 63
I1: maximum 44, but its questions add up to 42	no artefact can reach 100 % on I1	I1Q1 ladder changed in commit a170016 (2022)
I2Q7 = 5 $\times$ average vocabulary weight; highest weight 0.75	capped at 3.75 / 5; the MOD 3.0 namespace (w3id.org/mod#) earns nothing	I2.java, line 124; metadata.voc.config.json
EarthPortal is not in the known catalogues	F4Q1 / F4Q2 start one rung lower than on AgroPortal; F4Q2 is our 2nd largest loss	catalogs.config.json

Source read: github.com/agroportal/fairness, commit 24def6b (4 June 2026). All checks are reproducible from the notebooks and from `reference/`.

Four more things a curator should know

Presence, not correctness

A field counts as filled when it is not empty: `"N/A"` or `"to be done"` pass. Our audits share this limit for now.

Truncated percentage

$N = \lfloor \cdot \rfloor$, not rounded: OSO's 76.2 is shown as 76.

Scores are not dated

In portal mode, O'FAIRe serves cached results, with no date or submission id. On 17 September, OSO's panel showed other values (version IRI, DOI, counts) than its page.

We do not know which O'FAIRe commit runs on the EarthPortal server, nor how often its cache is refreshed.

Observations on the catalogue itself

- **5 metadata-only records look like tests**: `TEST`, `HG` (“Hello Go”), `SW`, `HIG` (“HI Go”), `GTD` (“GO TO DO”). Remove them?
- **Views of OSO**: `OSO-DCAT`, `OSO-VOID` and `OSO-SHACL` are now declared as views, which solves our earlier finding that `OSO-DCAT` shared OSO’s URI.
- **5 pull URLs** embed an API key of another OntoPortal instance (masked in all our outputs).

The last three points come from the snapshot of 28 September 2026 and must be checked again on the current one.

Questions I would like to discuss

For the EarthPortal team

1. Is the split **author** / **curator** / **portal** right?
2. Should **Location** stay Mandatory for uploaded files?
3. Who contacts the owners: authors (audits), and portals whose **keys** are exposed?
4. Can the **test records** be removed?

For the O'FAIRe team

5. Which **commit** runs on EarthPortal, and how often is the **cache** refreshed?
6. Could EarthPortal be listed in `catalogs.config.json`?
7. Where should our evidence go: GitHub issues on `agroportal/fairness`?

Environment and next steps

Environment: open and reproducible



github.com/EarthPortal/SA_Management

docs/: reports (LaTeX, PDF)
earthportal_sa_audit/: notebooks, tables, audits
one `.env`, one `requirements.txt`; MIT licence

The notebooks run on an HPC node of Area Science Park (Trieste)

Node	epyc006.hpc.rd.areasciencepark.it, partition EPYC (Slurm)
CPU	2 × AMD EPYC 7H12, 64 cores each (128 cores)
Memory	503 GiB, no GPU
OS	Fedora Linux 41
My session	Slurm job of 12 CPUs, 48 GB, 2 hours; VS Code through a tunnel
Python	3.13.9 in a virtual environment; pandas 2.3.3, fg-data-profiling 4.20.0

Reports are written in LaTeX on my PC (WSL Ubuntu, TeX Live, VS Code); the same git identity is used on both machines.

What comes next

Next weeks

- validate *who fixes* and the priorities with you
- re-check the catalogue observations on today's snapshot
- first contacts with authors, starting with P1

Then

- validation rules in the machine-readable guideline (value checks, not only presence)
- scheduled snapshots: a history of the scores
- compare with **FOOPS!**

The goal

A curation pipeline that **proposes** changes, which a curator **validates** in an admin interface (possibly with AI support).

Thank you for your attention

Merci beaucoup de votre attention · Grazie mille · Aksante saana

“All models are wrong, but some are useful.”

George E. P. Box

“Big goals, small steps.”

github.com/EarthPortal/SA_Management

Backup: the ten questions O'FAIRe does not implement

Question	Max	Text
F4Q3	8	Are the ontology libraries or repositories properly indexed by Web search engines?
I2Q3	3	If yes, does it include the minimum information for those terms?
I2Q5	7	If yes, are those alignments well represented and to unambiguous entities? . . .
I3Q1	20	Does an ontology provide qualified cross-references to external resources/databases?
I3Q2	6	If yes, are those cross-references well represented and to unambiguous entities?
R1Q3	7	How much of the ontology objects are described with labels?
R1Q4	6	How much of the ontology objects are defined using a text description?
R1Q5	6	How much ontology objects are defined using a property restriction or an equivalent class?
R1Q6	5	How much of the ontology objects provide provenance information with annotation properties?
R1.2Q5	2	Are the ontology latest changes documented?
Total	**70**	Every answer: "This question cannot be evaluated in the repository at this time."

Backup: the snapshots so far

Snapshot	Entries	Artefacts	Views	Metadata-only	OSO (reachable)
28 Sept 2026	91	70	16	5	93.9 %
2 Oct 2026	92	71	16	5	93.9 %
6 Oct 2026	92	71	16	5	94.4 %
8 Oct 2026	92	68	19	5	94.4 %

- by 2 October: `SCHEMA-ORG` (Schema.org) appears as an artefact
- between 6 and 8 October: `SCHEMA-ORG` is no longer public; `OSO-SHACL` appears as a view of OSO; `OSO-DCAT` and `OSO-VOID` become views of OSO
- median reachable score of the artefacts: 67.6 % (2 and 6 Oct), 67.85 % (8 Oct)

Tables in `earthportal_sa_audit/data/`, one set per snapshot date.